METFORMIN

Lactic acidosis

Hold for iodinated contrast

CHEMICAL NAME N,N-dimethylimidodicarbonimidic diamide (1,1-dimethylbiguanide hydrochloride)
CLASS Biguanide oral antidiabetic
TRADE NAMES Glucophage · Glucophage XR · Glucomet · Janumet (with sitagliptin)

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes — first-line (especially overweight).
- Prediabetes (prevention, selected).
- Polycystic ovary syndrome (insulin resistance, ovulation; off-label).
- Gestational diabetes (selected).

ADMINISTRATION

- **PO** with or after meals to reduce GI effects; start low, titrate slowly.
- XR tablets once daily with evening meal; swallow whole.
- Temporarily stop for iodinated contrast (eGFR <30 or AKI), surgery, acute illness with dehydration ("sick-day rules").

DOSAGES

- **Start:** 500 mg once daily with breakfast for 1 week, then 500 mg twice daily; increase to 1 g twice daily.
- **Max:** 2–3 g/day (XR max 2 g).
- **eGFR 30–45:** max 1 g/day; stop if eGFR <30.
- **Children ≥10 yr:** 500 mg daily, max 2 g/day.

PHARMACOKINETICS

- **Onset:** days; full effect 1–2 weeks. Bioavailability 50–60%.
- Not metabolised or protein-bound.
- Excreted unchanged in urine (accumulates in renal failure). **t½:** 4–9 h.

PHARMACODYNAMICS

- ↓ **hepatic gluconeogenesis** (AMPK activation), ↑ peripheral insulin sensitivity, ↓ intestinal glucose absorption.
- Does not cause hypoglycaemia alone; weight-neutral or loss; cardiovascular benefit.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Diarrhoea, nausea, abdominal pain, anorexia (dose-related, usually transient).
- Metallic taste.
- Flatulence.

ADVERSE EFFECTS (SERIOUS)

- Lactic acidosis (rare but serious); renal failure, hypoxia, sepsis, shock, liver failure, alcohol excess.
- Vitamin B12 deficiency (long-term → anaemia, neuropathy).
- Hypoglycaemia only when combined with insulin/sulfonylureas.

CONTRAINDICATIONS & CAUTIONS

- eGFR <30 mL/min.
- Acute conditions with tissue hypoxia: shock, sepsis, respiratory failure, acute MI, dehydration.
- Diabetic ketoacidosis; hepatic failure; alcohol intoxication.
- Iodinated contrast in at-risk patients (temporarily).

KEY INTERACTIONS

- **Iodinated contrast:** AKI → lactic acidosis.
- **Alcohol:** ↑ lactic acidosis.
- **Cimetidine, dolutegravir, ranolazine:** ↑ metformin levels.
- **Steroids, thiazides:** ↑ glucose.

NURSING MANAGEMENT

- Check eGFR at baseline and at least yearly (more often if declining).
- Give with meals.
- Hold before contrast/surgery per protocol and restart after 48 h if renal function stable.
- Monitor HbA1c, B12 (annually), GI tolerance.
- Watch for signs of lactic acidosis (malaise, myalgia, rapid breathing, abdominal pain).

PATIENT EDUCATION

- Take with meals; stomach upset usually settles in a few weeks.
- Stop temporarily if vomiting, diarrhoea or unable to drink — restart when well (sick-day rules).
- Limit alcohol.
- Tell staff you take metformin before X-ray dye or surgery.

KEY POINTS

- First-line for type 2 diabetes; no hypos alone.
- Stop if eGFR <30 or acute illness/contrast.
- Start 500 mg, titrate; take with food.
- Lactic acidosis; B12 deficiency.

SOURCES CHECKED **[597]** Metformin 1000 mg film-coated tablets (UK SmPC) **[598]** Metformin, oral (US label) **[599]** Metformin, oral (US label, REMEDYREPACK INC)
Full references p. 432–457